

Errors in Variables

Every earlier chapter takes the regressors as recorded without error. Often they are not: one clinic reading stands in for long-run blood pressure, a questionnaire for usual diet. The scientific model concerns the true value X , but the data contain a noisy W , and least squares on W estimates the best linear predictor from what was recorded. This chapter measures the gap and closes it. Classical error flattens a slope by a factor no sample size removes, makes correctly measured covariates look important, and in the normal model leaves the slope unidentified. Information from outside the regression rescues the analysis: a known error variance, repeated measurements, a known variance ratio, an instrument, or validation data. The moment correction, instrumental variables, regression calibration and SIMEX turn that information into estimates, each with its own conditions and costs.

What you need. [Chapter 6](#) (the population projection, Theorem 6.11.1 and Proposition 6.11.3, and Theorem 6.3.6). [Chapter 5](#) (moments of least squares under departures, Proposition 5.5.7). [Chapter 14](#) (random regressors, Theorem 14.1.2, and the limit theorems of [Section 14.4](#)). From this part, [Chapter 19](#) (omitted-variable bias, Theorem 19.1.2), [Chapter 21](#) (sandwich covariance estimators, Theorem 21.4.2) and [Chapter 23](#) (the pairs bootstrap, Definition 23.1.1).

24.-1.1 Roadmap

- [Attenuation bias](#)
- [The classical measurement error model](#)
- [Reliability and moment correction](#)
- [Instrumental variables](#)
- [Regression calibration and SIMEX](#)
- [Summary and notes](#)